

# NIKHIL BISHT

Delhi, India

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## Education

**Maharaja Surajmal Institute of Technology**

**Aug. 2024 – Jun. 2028 (Expected)**

*Bachelor of Technology in Information Technology (CGPA: 9.451/10.0)*

*New Delhi, India*

## Technical Skills

**Languages:** Python, C, C++, Java, JavaScript, SQL

**AI/ML:** ML, LLMs, Attention Networks, Deep Learning, NLP, Computer Vision, OpenCV

**Frameworks:** PyTorch, TensorFlow, LangChain, LangGraph, YOLOv8, FastAPI, Django, Streamlit

**Web Development:** React, Node.js, RESTful APIs, HTML, CSS

**Tools & DevOps:** Docker, Postman, Hugging Face, Vercel, Render, GitHub

## Research & Publications

**ANN for Fashion Product Classification | NCERTES**

**Feb 2025**

- Engineered a high-precision Artificial Neural Network (ANN) architecture for multi-class fashion imagery classification, achieving state-of-the-art accuracy on bench-marked datasets.
- Co-Authored and presented the research paper at the NCERTES conference, detailing model optimization and data augmentation strategies.

## Projects

**Mistral-7B LLM: Full-Stack Architecture Implementation | PyTorch, LLMs**

**Feb. 2026**

- Successfully reconstructed the **Mistral architecture from scratch**, implementing key features like Sliding Window Attention (SWA) and Grouped-Query Attention (GQA) for efficient inference.
- Integrated Rotary Positional Embeddings (RoPE) and RMSNorm to stabilize training and improve long-context understanding; verified performance through rigorous layer-by-layer weight initialization tests.

**English-to-Hindi Custom Transformer | PyTorch, NLP, Streamlit**

**Mar. 2026**

- Developed a **complete Encoder-Decoder Transformer from scratch** optimized for English-Hindi translation, featuring 30M trainable parameters and custom attention heads.
- Built a specialized BPE tokenizer and trained the model on 1M+ parallel sentence pairs, achieving significant BLEU score improvements over standard baseline models.

**AI vs. Real Image Detection (DALL-E Focus) | ResNet-34, PyTorch, CV**

**Apr. 2026**

- Constructed a hybrid deep learning model for detecting synthetic imagery from LLMs by building a **ResNet-34 backbone from scratch** to extract high-resolution spatial features.
- Appended Transformer decoder layers to analyze global pixel dependencies, identifying subtle artifacts unique to diffusion models and modern LLM-based image generators.

**Real-Time Object Tracker (YOLOv8m) | OpenCV, YOLOv8, Streamlit**

**Jan. 2026**

- Engineered a robust multi-camera pipeline using YOLOv8m for simultaneous object detection and tracking with unique ID persistence across occlusions and high-resolution streams.

**Flappy AI (Reinforcement Learning) | Gymnasium, DQN**

**Mar. 2026**

- Developed and trained an autonomous agent using **Deep Q-Networks (DQN)** built from scratch to master the Flappy Bird environment. Optimized performance via Epsilon-Greedy strategy and Experience Replay.

## Competitive Programming

**Problem Solving:** Solved **1000+** DSA questions across platforms like LeetCode, Codeforces, and GFG.

**Skillset:** Strong proficiency in Dynamic Programming, Graph Theory, and Advanced Data Structures in C++.

## Certifications

**Machine Learning Specialization | Andrew Ng (Stanford/DeepLearning.AI)**

**Complete Data Science | Krish Naik (iNeuron)**